NA_2024_Lecture25

(1)

Iterative techniques to solve $Ax = b$

In this chapter we will discuss numerical methods to solve a linear system of equations $Ax=b$, numerically. Before we delve into numerical methods, we need to know some priliminory concepts related to vectors and matrices, such as norm, convergence, etc.

Various types of norms are defined for vectors, however, we will stick to only following two kinds of norms.

l_2 norm

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in R^n \qquad \|x\|_2 = \left[\sum_{i=1}^{n} |x_i|^2\right]^{1/2}$$

eg
$$x = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad \|x\|_2 = \sqrt{(1)^2+(-1)^2+(0)^2}$$

l_∞ norm

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in R^n, \qquad \|x\|_\infty = \max_{1\le i \le n} |x_i|$$

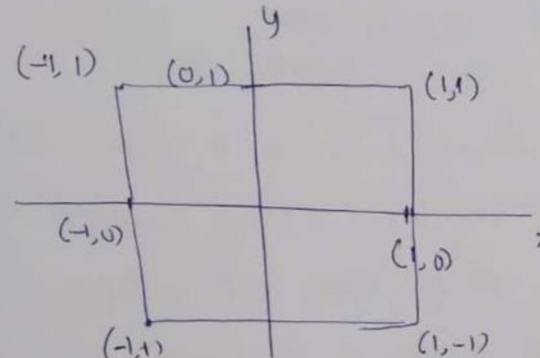
eg
$$X = \begin{bmatrix} 2 \\ -4 \\ 5 \end{bmatrix}, \qquad \|x\|_\infty = 5.$$

Generally, norm $\| \ \|$ is a function from $\mathbb{R}^n \longrightarrow R$, with following properties.

1. $\|x\| \geq 0$, $x \in R^n$,
2. $\|\alpha x\| = |\alpha| \, \|x\|$, $x \in R^n$ & $\alpha \in R$,
3. $\|x+y\| \leq \|x\| + \|y\|$ $x, y \in R^n$,
4. $\|x\| = 0$ if and only if $x = 0$.

In every norm space the set of unit vector ($\|x\|=1$) is an importont port of it. Let us explore how they look like in both l_∞ & l_2 norms.

$\|x\|_\infty = 1$ in R^2



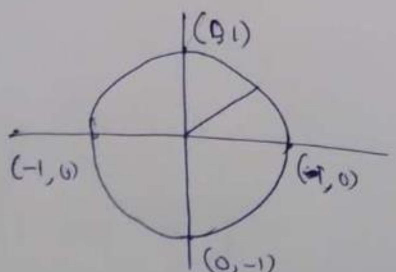
$\|x\|_\infty = \max\{|x_1|, |x_2|\} = 1$

This implies that one of $|x_1|$ or $|x_2|$ must be 1.

All vectors on the boundary of the squore satisfy $\|x\|_\infty = 1$. ~~Moreover,~~ Moreover, set of vectors $\|x\|_\infty \le 1$ is called unit sphere.

$\|x\|_2 = 1$ in R^2

$\|x\|_2 = 1 \Rightarrow \sqrt{x_1^2 + x_2^2} = 1 \Rightarrow \boxed{x_1^2 + x_2^2 = 1}$ unit circle centered at $(0,0)$



All vectors on the circumference of unit circle satisfy $\|x\|_2 = 1$.

In this case $\|x\|_2 \le 1$ unit sphere is a disck.

Having defined set of unit vectors in each norm, we are in position to define norm of a matrix.

Norm of Matrix

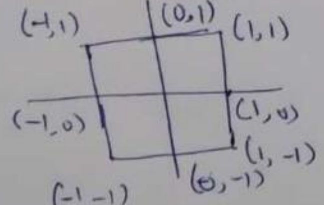
If A is $m \times m$ matrix its norm is defined by

$$\|A\| = \max_{\|x\|=1} \|Ax\| \quad \text{—} (1)$$

The norm $\| \; \|$ here could be any of l_∞ & l_2 norm. It depends which norm we use on the right hand side for vectors. The norm of A represents the maximum impact on the length of x when A acts on x. In particular, in definition (1), we transform entire $\|x\|=1$ using A and find the maximum length change.

Example

Consider $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$. Let us calculate the impact of A on $\|x\|_\infty = 1$.



$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $Ax = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\|Ax\|_\infty = 2$

$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $Ax = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\|Ax\|_\infty = 3$

$x = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $Ax = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\|Ax\|_\infty = 3$

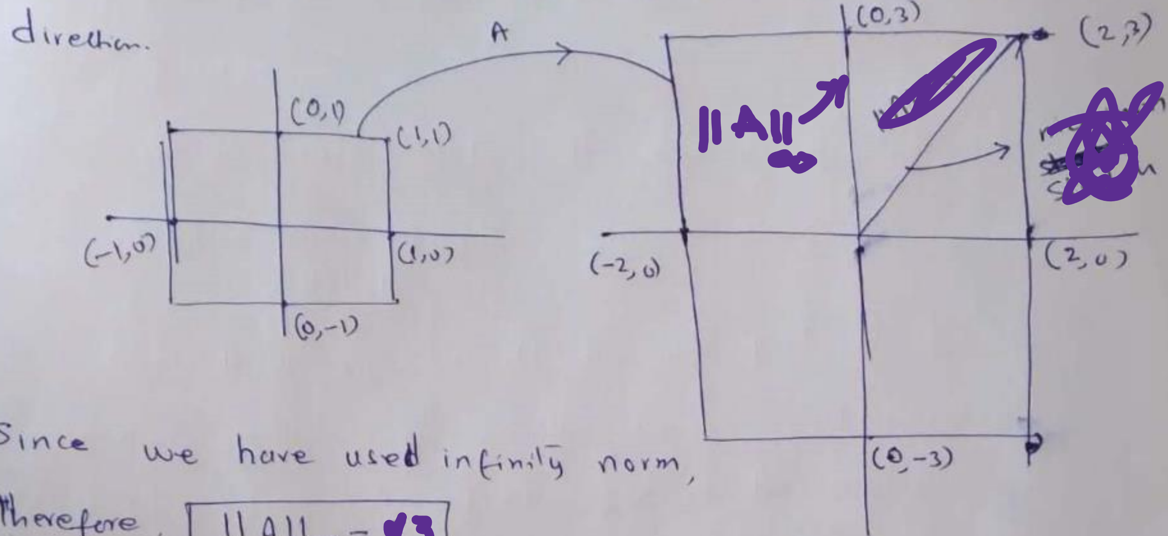
⋮

Generally

$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $Ax = \begin{bmatrix} 2x_1 \\ 3x_2 \end{bmatrix}$, $\|Ax\|_\infty = \max\{|2x_1|, |3x_2|\}$

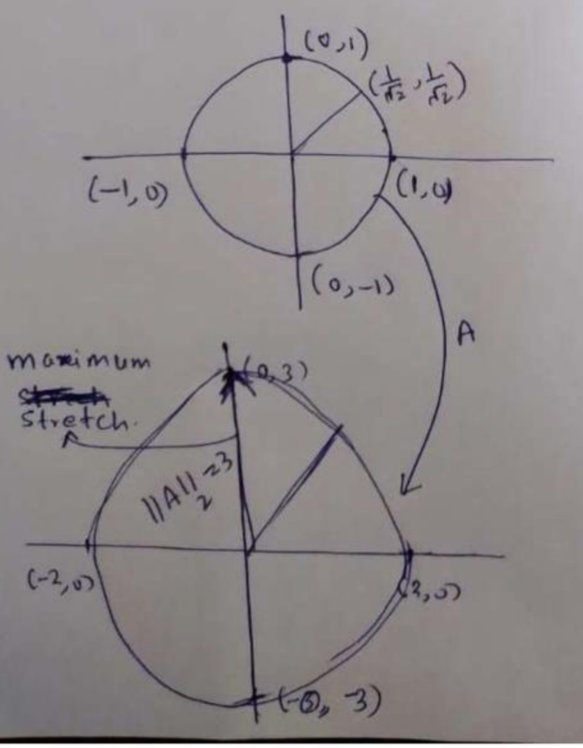
$\|Ax\|_\infty$ would be maximum if both x_1 and x_2 are maximum. In this case they are $x_1 = 1$, $x_2 = 1$, therefore $\boxed{\|A\| = 3}$

So, basically A has mapped $\|x\|_\infty = 1$ to a rectangle by streching it 2 units in x-direction and 3 units in y direction.



Since we have used infinity norm, therefore, $\boxed{\|A\|_\infty = 3}$

Now, we calculate $\|A\|_2$, ie., we investigate maximum impact of A on $\|x\|_2=1$ (unit circle) in R^2.



$x = \begin{bmatrix}1\\0\end{bmatrix}$, $Ax = \begin{bmatrix}2\\0\end{bmatrix}$, $\|Ax\|_2 = \sqrt{(2)^2+(0)^2} = 2$

~~$x = \begin{bmatrix}1\\1\end{bmatrix}$, $Ax = \begin{bmatrix}2\\3\end{bmatrix}$, $\|Ax\|_2 = \sqrt{(2)^2+(3)^2} = 5$~~

$x = \begin{bmatrix}0\\1\end{bmatrix}$, $Ax = \begin{bmatrix}0\\3\end{bmatrix}$, $\|Ax\|_2 = \sqrt{(0)^2+(3)^2} = 3$

$x = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$, $Ax = \begin{bmatrix}2/\sqrt{2}\\3/\sqrt{2}\end{bmatrix}$, $\|Ax\|_2 = \sqrt{(\frac{2}{\sqrt{2}})^2+(\frac{3}{\sqrt{2}})^2}$

$= \sqrt{\frac{4}{2}+\frac{9}{2}} = \sqrt{13}$

$= 2.55$

$x = \begin{bmatrix}x_1\\x_2\end{bmatrix}$, $Ax = \begin{bmatrix}2x_1\\3x_2\end{bmatrix} \Rightarrow \|Ax\|_2 = \sqrt{(2x_1)^2+(3x_2)^2}$

$= \sqrt{4x_1^2+3x_2^2}$

This suggests that maximum ~~$\|Ax\|_2$~~ value of $\|Ax\|_2 = 3$

Therefore $\|A\|_2 = 3$.